

Feasibility Analysis of Transparent Agrivoltaic Systems in Farms and Ranches

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1. Abstract

Agrivoltaics, the dual use of land for agriculture and solar energy, has a large potential energy output because 52% of US soil is used for agriculture. However, large-scale implementation of agrivoltaics is difficult because their stationary nature prevents large machinery to be used within fields, deteriorates soil health, and blocks sunlight exposure. To combat this issue this research focused on design and analysis of the feasibility of a transparent agrivoltaic structure implementation in farms and ranches. The design was implemented using rigid origami techniques, resulting in a fully functional, collapsible origami solar array and a complimentary housing tower for it to collapse into. Unfortunately, there was not enough time to fully design the power source and other details, but promising ideas have been drafted for these concerns. To model the application of the transparent agrivoltaic systems, Agricultural Production Systems Simulation (APSIM) and customized weather data was used to mimic the effects of agrivoltaic panels on crop growth. Homer Pro software was utilized to model energy production based on the specification of Brite Solar's transparent PV panels and the group design. Both models allowed simulated comparison between implementing transparent, generic, and no agrivoltaics. Model analytics show that the transparent PV's are viable for energy production and have minimal effects on crop yield.

2. Introduction

Sustainability in farming and energy is a great concern for the US. Surprisingly, they go hand in hand through agrivoltaics, targeting two issues with a single system. By building solar structures that sit within agricultural land, it is possible to add more renewable energy to the grid while making farms more self-sufficient and cost-effective. The economic benefits are possible because agrivoltaic structures have the potential to shield crops from the harmful effects of increasing temperature due to climate change. This could mean

reducing water consumption or allowing for the growth of crops that cannot grow under too much sunlight exposure.

However, there are bound to be issues with this relatively new technology. One is obviously the upfront and maintenance costs but the main issue we are concerned with is scalability. Most integrations of agrivoltaic systems currently exist on small scale farms that mostly use manual labor to conduct agricultural operations. This allows for a satisfactory population of solar arrays within a field because they can be more densely populated.

In large farms and ranches, industrial machinery is used, which creates big spacing constraints between agrivoltaic structures.

Finally, agrivoltaic structures make the area underneath them unsuitable for crop growth because the sun becomes completely blocked.

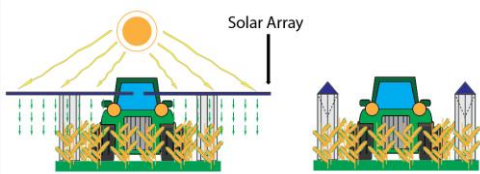
Our solution to these problems targeted the aforementioned shading effect on farms, and the scalability issue. This was done through the introduction of transparent solar panels and origami to make a collapsible, transparent solar array.

Data analysis was also conducted on the potential effects of the design through computer simulations. The setting of our analysis was corn farms in West Palm Beach, Florida. This location was chosen for its prominence in corn production, but also for its similarity to southern states with the harshest climate conditions where the benefits of agrivoltaics may be most evident.

3. Mechanical Design



Above is the “Solar Bloom”, the original concept model for our design idea. As mentioned before, it is a collapsible, transparent solar array.



The figure above depicts how this design would combat the issues of scalability, and the decrease of crop yield under the array. The folding of the array allows for the minimum spacing of agrivoltaic structures to be defined by the base of their towers rather than the reach of the expanded solar array. The transparent panels will allow for some light to pass through the array, allowing for growth of corn beneath the structure.

It is to be noted that due to the time constraints, that priority within the design was put into actualizing new ideas such as the origami array, mathematical modeling of the array, and a resultant folding automatic closing and encapsulation structure rather than engineering details concerning design for manufacturing, material selection, and critical fasteners.

Therefore, the design is in no means fully finished, we like to believe that the main innovation within the design has been achieved.

3.1 Part Selection

This section concerns the selection of transparent solar panels, and hinges used within the solar array.

Transparent solar panels are a relatively new technology, being only about a decade old. They are not very favorable in most industries requiring solar panels because their efficiency lies within the 5% range while modern opaque solar panels sit around 20-30% efficiency. So when choosing our solar panels, one of our criteria was having a high efficiency, but also unfortunately the responsiveness of certain companies to our inquiries as we also required that these solar panels have a customizable shape. Unfortunately, only one company really responded to our emails, which was Brite Solar.

During our conversations with their CEO, Dr. Nikos Kanopoulos, we were able to confirm that their panels are customizable in shape, and that they have a

model of solar panel specifically designed for growing corn, as their company also has a heavy interest in agrivoltaics. This is something that we have had to take their word for, as they claim to have third party research done supporting that corn growth underneath the BSG-150/69-F is essentially the same as if the corn is exposed to unfiltered sunlight.

As for the hinge selection, we had two main criteria for the type of hinge. We wanted the hinge to be robust, meaning low part count and compact design, and a circular swing path. These criteria are things that will be expanded upon in section 3.2.1.3, Hinge and Panel Accommodation.

We considered three main hinge types, the SOSS invisible hinge, surface mount butt hinges, and mortise butt hinges, which are depicted in the listed order below.



The SOSS invisible hinge is, as the name suggests, invisible to a certain extent. With its unique design, the hinge engulfs itself when closing, becoming effectively invisible. This is something that would be aesthetically pleasing, but lots of problems for our design. As seen in the image above, the numerous hinge components are very exposed when open, making this unfit in our “robust” criteria. Another deal breaker is that the hinges do not have a circular swing path.

As for the surface mount butt hinge, it is robust and has a circular swing path. But when considering its close relative the mortise butt hinge, it is simply a worse option because it has to sit on top of whatever frame or door it holds. This is unideal because as it will be explained later on, added thickness reduces the total solar area within our area.

The chosen hinge is a “Mortise-Mount Entry Door Template Hinge” from McMaster with the part

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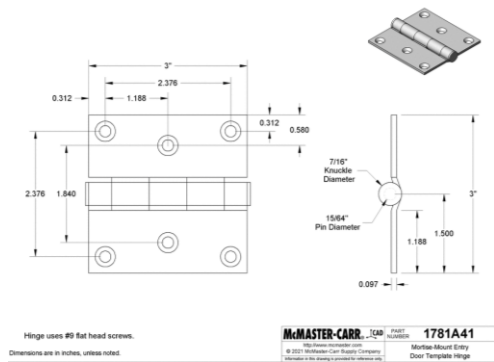
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When considering the dimensions of the hinge, smaller is generally better so that unnecessary support structure can be avoided in the array. However, it turns out that the BSG-150/69-F had dimensions that required more support structures than anticipated, making this hinge satisfactory in terms of dimensions.

3.2 Creation of the Solar Array

Creation of the Solar Array was something that took most of the 9 weeks of the program because it utilized many new techniques all within the practice of rigid origami.

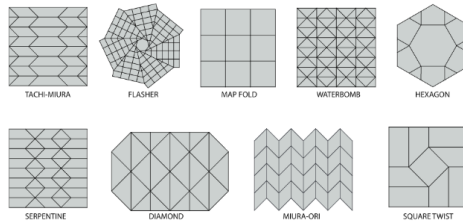
3.2.1 Rigid Origami

Rigid origami is the practice of creating origami structures out of non-zero thickness, inflexible materials. It has many applications to solar structures in space for its ability to store large arrays, but it is also making its way around the entire engineering community. From nano injectors to meta-materials, it has lots of untapped potential, including agrivoltaics.

Rigid origami follows three main steps: choosing a crease pattern, thickness accommodation, and hinge accommodation. However for the unique purposes of our design, we will also had to accommodate for the panel thickness, which will be be combined with the hinge accommodation.

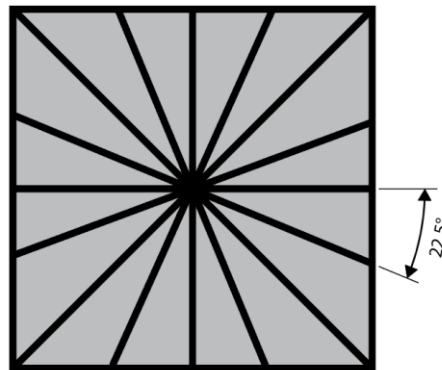
3.2.1.1 Crease Pattern Design

A crease pattern in both traditional origami and rigid origami is what you get when you unfold an origami pattern and trace out the crease markings.



Seen above are many popular crease patterns within engineering. Notice that a common trait between these patterns is repetition. Repetition in origami is a sign of compatibility which is of great interest to engineers.

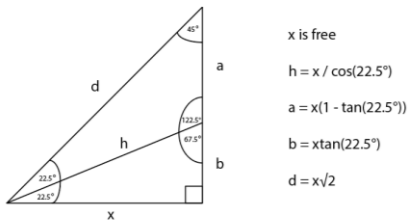
Another defining characteristic of crease patterns is their degree of vertex. This is defined by how many creases intersect at a single point. For example, the tachi-miura fold would have a degree 4 vertex. This is generally a tell of how difficult the design is to actualize in rigid origami because it is an early tell of how much interference one might get when transforming the origami from paper to thick materials.



Shown above is our crease pattern. It is an original design as far as we can tell. It is a square

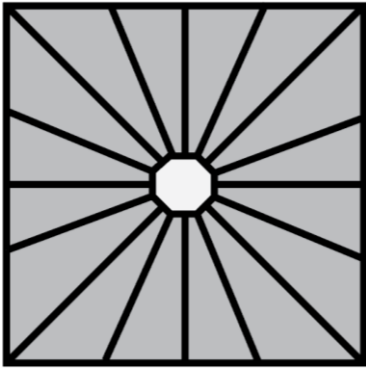
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sectioned off into 16 sections by a central angle of 22.5 degrees.



In this “crease-pattern” form, it only has two unique shapes as shown in the mathematical diagram above, but this is something that will change as the design progresses.

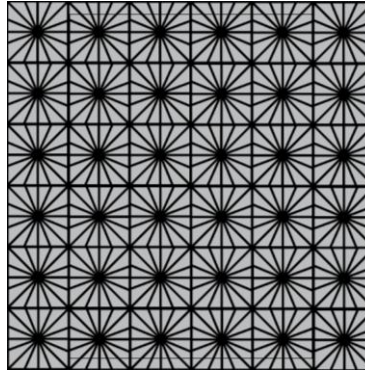
Using the characteristics of crease patterns mentioned before, this has quite a repetitive pattern, which is good, but would be very difficult to implement due to its 16 degree of vertex. However for the purposes of our design, this is actually deceiving as the center of the crease pattern is not crucial to the key function of our array.



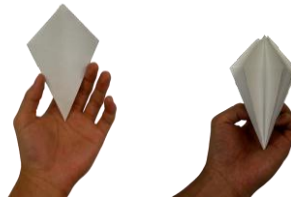
As seen here, without the center, individual pieces are still connected to each other, and solar panels can still comfortably fit within each section of the array.

This is something that would be more important to consider if the crease pattern became a repeated unit itself, as shown below, since then the details about how the pieces of the array connect and intertwine become a little more sensitive. But this is just something to note as

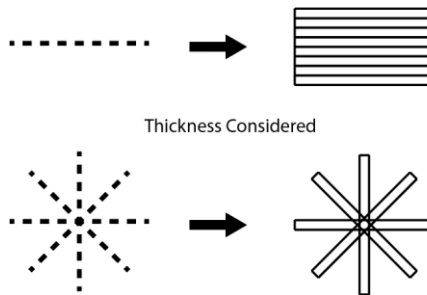
an interesting application of our crease pattern.



When actually folding this pattern there are two main ways to go: folding it “circularly” and folding it flat which are shown respectively.



Below, is what these two options would look like when considered from a top down view.



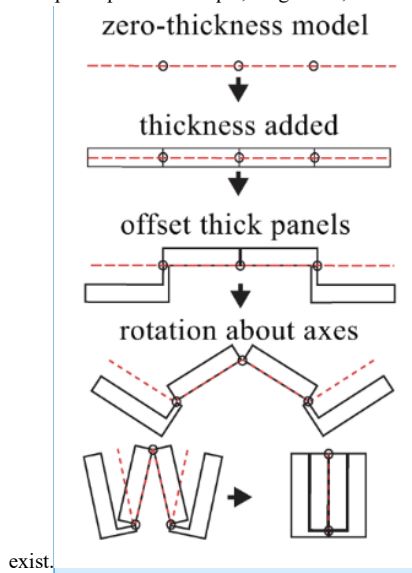
At first, the flat option was favorable for its extreme compactness, but was deceiving since the real compactability is compromised by the thickness of the panels. Additionally, folding the array in this manner from an open state takes many more steps in many

directions, which is greatly unfavorable when considering this design is supposed to be automatically opened and closed.

The “circular” folding option is named so because the way it folds creates a constant circular symmetry, which will allow for a single closing and opening motion.

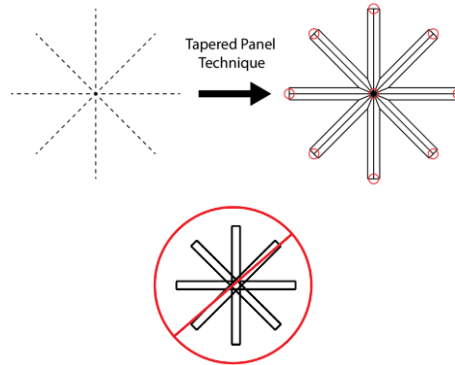
3.2.1.2 Thickness Accommodation

Hinge accommodation is the first step that takes traditional origami into useful engineering solutions. As the name may suggest, this is where we first consider the thickness of our material, and then accommodate it to get rid of any resultant interference. Many techniques have become popular enough to gain an official name, like the offset panel technique shown below, and many others like the tapered panel technique, hinge shift, double hinge, etc



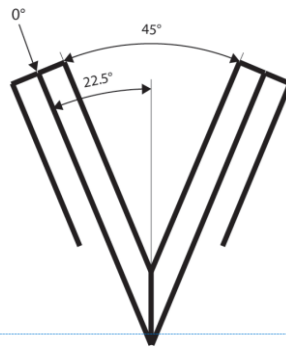
The image above shows BYU’s suggestion when taking on this step. Consider the zero-thickness model from an open view, adding thickness, adding your accommodation technique, and rotating about your axis to confirm that the accommodation works.

In our experience, this step turned out to be more intuitive than anything, but what did help that is probably applicable to other designs is taking the steps offered by BYU somewhat backwards.



When considering the design beginning with the crease pattern folded with thickness added, it's easier to see where interference of the panels actually occurs. The small red circles point out where the folds exist. The crossed out image shows the unaccommodated version of the array.

The technique used is the tapered panel technique. As the name suggests, this technique tapers the panels to solve interference.



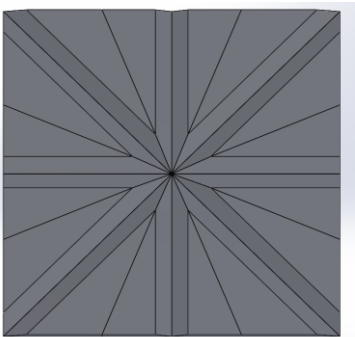
When defining the taper, it is important to note some additional characteristics of the origami as it folds. As shown in the figure depicting the tapered panel technique, it can be seen that the general shape of the

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folded array is a circle divided into 8 equal parts, this would mean that the angle between these parts is 45 degrees, as labeled in the zoomed in image above. This means that for the tapers to sit flush when closed, the angle on the tapers must be 22.5 to add up to 45 degrees in each section.

It is also important to note the zero degree label in the image above because this differentiates two unique folds within this design. It is called the zero degree fold because the panels fold to zero degrees there. The other fold is the 45 degree fold between the tapered ends of the array. From now on, these names will reference this distinction.

Using the information accumulated from the past two subsections, a real proof of concept was created using a 3D printer, as shown below.



It was mentioned in this paper that the number of unique pieces would increase as the design progresses. This is one of those progressions. Originally, with just the no-thickness version of the design, there were only 2 unique shapes. But now, as seen above, these two shapes have mirrored versions that are unique because the 3-dimensional versions have tapers on single edges. This accounts for a total of 4 unique shapes at this point.



This proof of concept has a total square edge length of 10 cm, with a panel thickness of .2 cm.

3.2.1.3 Hinge and Panel Accommodation

Hinge Accommodation is similar to thickness accommodation in the way that we must consider the space that a hinge may take up, and account for interference.

However, for the application of our rigid origami design, we also have to consider how the frames change considering the solar panel. The panel thickness itself actually has a lot of importance to how much solar panel area we get out of our design.



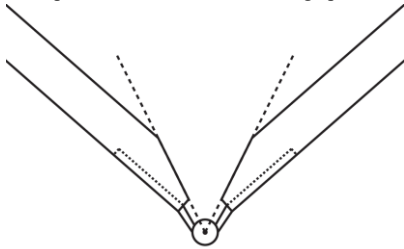
The image above depicts all of our basic constraints when making the design. The dashed line is a part of the tapered end of an array piece that must be removed to replace with the hinge selected in the previous sections.

The chosen solar panel has a thickness of 30 mm and a buffer of .5 inches around it to frame it. This is not very ideal because through the tangent relationship with the 22.5 degree taper, the horizontal length of this taper becomes 4.058 inches including the dashed sections. This extra long support section cuts into our available solar panel area. This can be solved by opting for a slimmer panel choice, however it was stated before that the BSG-150/69-F was chosen for the company's claim that it is the most ideal transparent solar panel for corn.

The horizontal length of the dotted area that is depleted from the original taper shape is determined by

the dimensions of the hinge. More specifically, this removed section is the length of the hinge from the center of its pin to the beginning of its leaf, which is .312 inches.

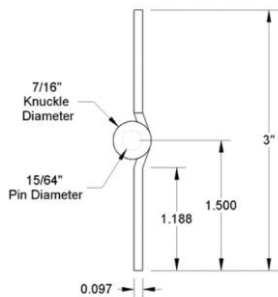
Another constraint when implementing the hinges is making sure that the correct faces of the frame line up to the center axis of the hinge pin.



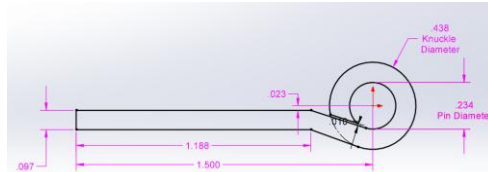
As seen in the image above, this is because we want the tapered ends of the panels to meet each other when folding to create the 45 degree angle discussed in the thickness accommodation section. The same goes for the 0 degree fold, although it is clearer how it works for this fold because it is similar to the opening and closing of a door.

This actually requires very specific dimensions for the hinge, which was not met by the original design. Therefore, slight alterations to the hinges were made to meet these requirements.

It is to be noted that this may potentially change the load capacity listed by the manufacturer, although there is reason to believe that load capacity may not change at all because the dimension changes are very minimal.

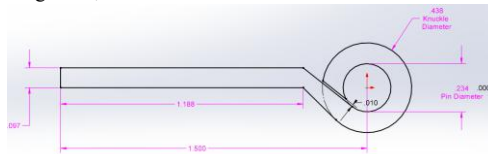


Seen above is the original drawing for the selected hinge. Changes to the only concern the dimension between the frame leaf faces and the parallel line drawn through the center of the hinge pin.

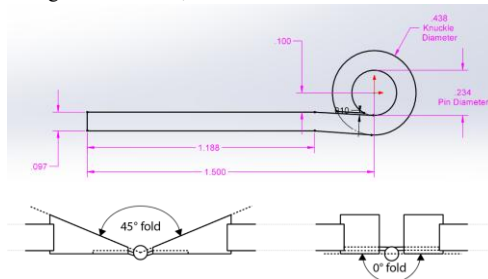


As shown above, in the original design, this dimension is .023 inches.

For the 45 degree fold hinge, this dimension was changed to .097. In other words, the bottom of the leaf was made collinear with the horizontal line through the hinge axis, as shown below.



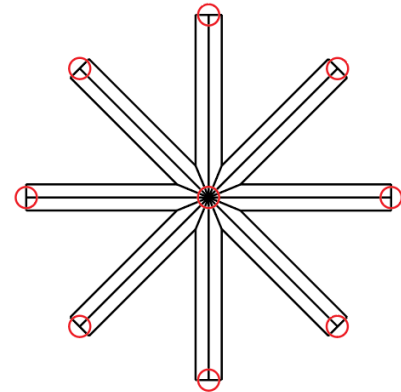
As for the 0 degree fold hinge, this dimension was changed to .1 inches, as shown below.



Shown above is an accurate depiction of the frame pieces connected to the hinges. One may notice that the .5 buffer is unbalanced, with most of the thickness being on one side of the panel than the other. This is because the side that has more material actually takes the force of the framed panel when the panel is open, requiring more support material. The other side merely needs to hold the array in place. The exact material requirements for this could be optimized and possibly reduce the dimensions of the supporting frame, and

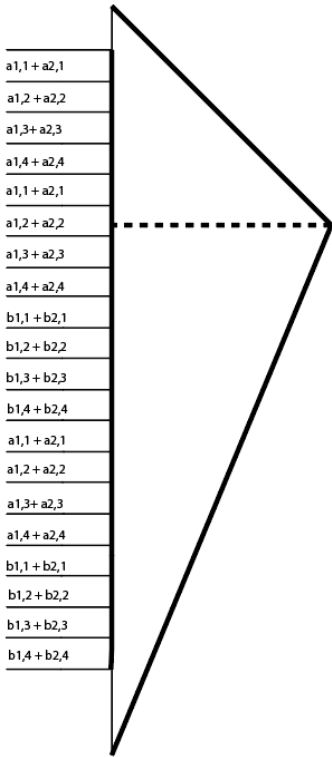
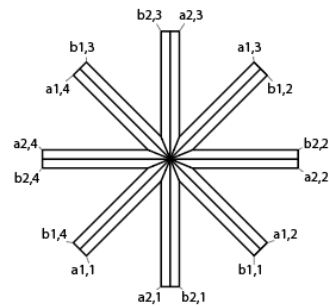
ultimately create more area for solar panels, but this was not done due to a lack of time.

Finally, we must be concerned with the exact placement of the hinges throughout the frame pieces.



In the image above, red circles represent the areas where hinges will exist and connect frame pieces. The circles that only contain two frame pieces is where all the 0 degree folds occur. For the hinge placement here, there is not much to think about since they will not interfere with any other hinges.

However, the red circle in the center represents all the 45 degree folds. This is an issue because that means that there is a lot of potential for hinge interference as they all become collinear when the array closes.



The solution lies within the two images above. Because of this unique problem, hinge placement must be carefully calculated and individualized for each panel. This is the last diversification between frame shapes, resulting in 16 unique frames.

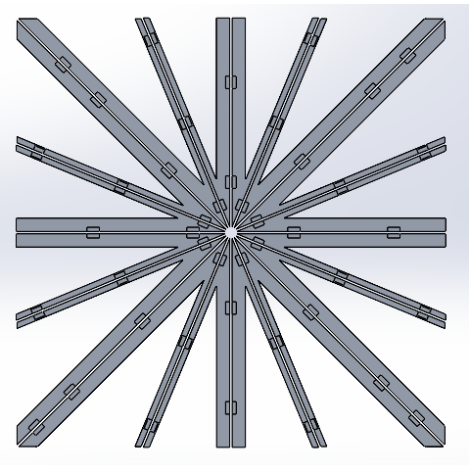
Each frame now has a unique name, as seen in the first image, and every 45 degree fold between the frames is represented in the second image.

We like to call this the periodic hinge technique because hinges between the frames periodically repeat as we go down the folded array.

Pieces that start with “a” are the larger of the two original pieces in the crease pattern, and “b” refers to the smaller counterpart.

The diagram used to track the hinge placement uses the shape of an “a” panel because below the dashed line is the same exact shape as a “b” panel.

One may also notice that resultantly, above the dashed line, only “a” hinges exist because there would be possibility of “b” hinges there. This allows for “a” panels to get three hinges each through the 45 degree fold, while “b” panels get 2 each through the 45 degree fold. The periodic placement is done in order for the hinges to be spread out on the frames, even if they sort of exist in different areas.



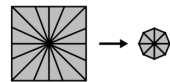
Above is the real, resultant model of the 16 unique frame pieces all assembled in the correct order.

3.2.2 Array Specifications

Here are some specifications about the completed array shown above.

The array spans 10 ft by 10 ft, and is 1.681 inches thick.

Upon closing the array, a top down view of the square footage reduces by 90%, as seen below.



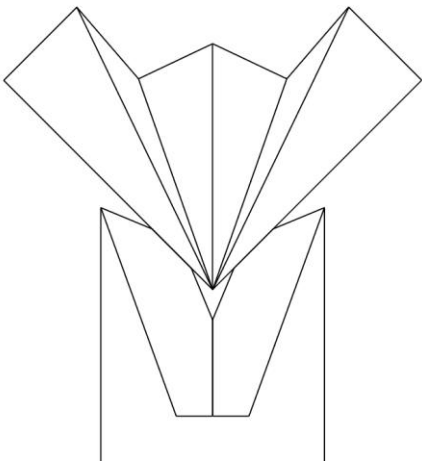
60% of the 100 square feet is solar panel.

The array frame is speculatively made out of 6061 T6 Aluminum as a commonly known lightweight, but strong and durable material. Stress analysis and loading calculations along with a materials study would give us a more certain answer on the best material but given time restraints, this material was applied for now.

With that being said, the array would be 470.25 pounds of 6061 T6 Aluminum, 161.47 pounds of solar panel, and 1.89 pounds of hinge for a total weight of 633.61 pounds.

3.3 Creation of the Tower Structure

The idea created to store the array also serves the dual purpose of folding the array.

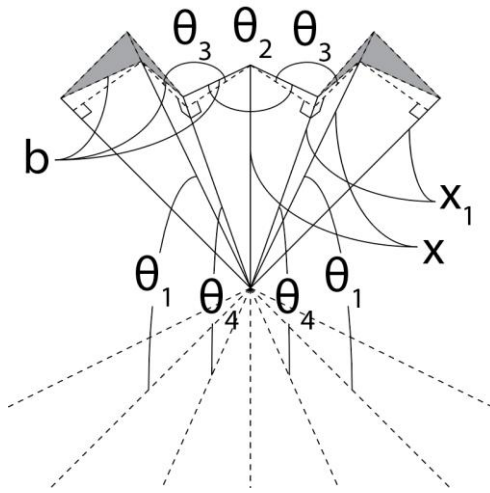


As seen above, the array sits partially folded inside the top of a tower structure. We call this a “multi-stage” guide rail system. The stages refer to the small sections within the tower at different angles. In the image, one of them is in contact with the array. The idea is that the tower can guide the array open and close by mimicking the shape of the array at various levels of opening.

3.3.1 Mathematical Model of the Solar Array

This was attempted at first through a Solidworks assembly animation. However, we learned that this cannot be done easily because the array cannot be mated properly due to the circular nature of the pieces. The fact that each mate is connected to every other mate creates something called a circular dependency, which is something that the software despises greatly. It is to be noted that we believe this to be possible through Solidworks, it would just require that mates are made to separate entities like planes rather than each other and then carefully constrained to mimic the desired movement.

However, it was deemed more valuable to achieve this through math because it would be formulaic, robust, and reliable.



The image above defines all the basic dimensions necessary to know how the array moves through space when opening. Any additional variables used during the derivation will be derived from these dimensions. The shaded areas are there to show how “a” panels are really “b” panels when disregarding the shaded areas, and it will be treated as so when deriving for easier variable naming use. Additionally, it is important to note exactly how

angles 2 and 3 are defined. Since angle 3 is a measure of the panels connected by the 0-degree fold hinges represented by x_1 , it is crucial that the lines used to determine angle 3 are perpendicular to x_1 . This would not be a problem if angle 3 was defined by the angle between the planes holding the two faces, but this was not how the derivations were completed. This is not as much as an issue for angle 2 because dimension b is already perpendicular to x , satisfying this condition. The goal was to relate angle 1 to all the other angles.

Another helpful constraint to keep in mind is that any dimensions of x or x_1 will always stay within the vertical plane that goes through the lines containing x and x_1 when the array is open. This is because this is the behavior that the array exhibits when folding “circularly” as stated in crease pattern selection. That is what the dashed lines represent. As a result, these dashed lines sit beneath x and x_1 as their potential projections onto the flat plane, which will be very useful for our derivation, and also help to define angle 1 and 4 a little more clearly.

Understanding the angle definitions is a little easier for angle 2 and 3. Angle 2 represents the angle between the faces of the 45 degree fold so naturally this angle ranges from 180 degrees when the array is open, to 45 degrees when the array is closed.

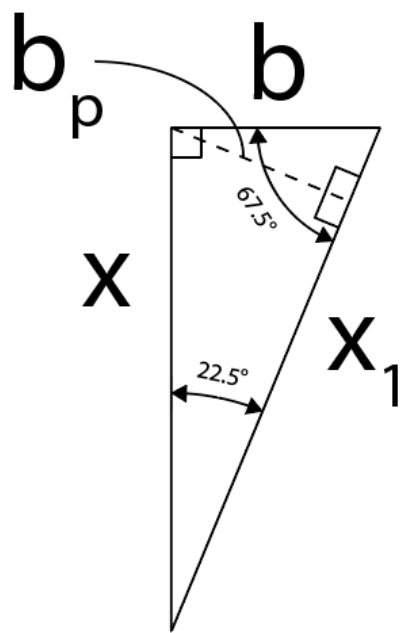
Angle 3 is the 0-degree fold, so it ranges from 180 degrees and 0 degrees.

The first angle that can be derived from angle 1 is angle 3.

This being said, we can use the projection lengths l , and known angles between them to define c .

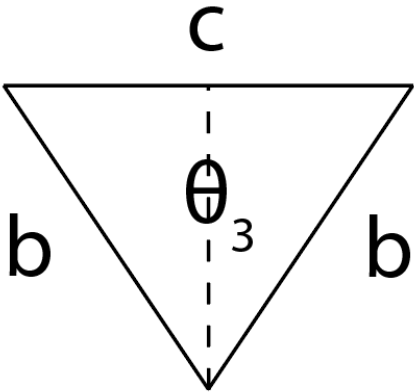
$$\begin{aligned} l &= x \cos(\theta_1) \\ c &= 2l \sin(22.5^\circ) \end{aligned}$$

The only thing left to do is to define b_p , and then use b_p and c to define angle 3.



$$b_p = x \sin(22.5^\circ)$$

“ B_p ” is defined rather simply.



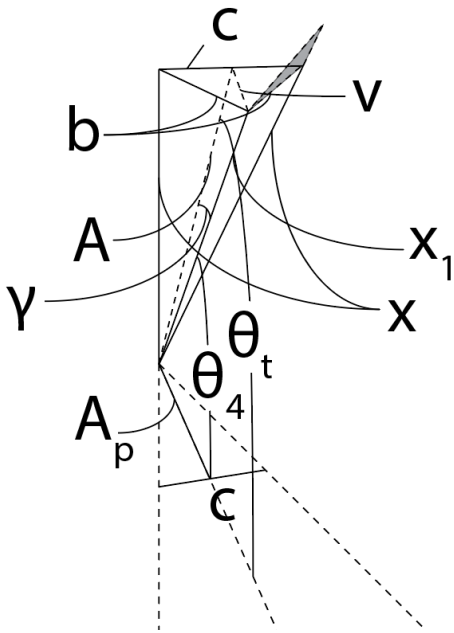
Finally, angle 3 is defined as follows.

$$\theta_3 = 2 \arcsin\left(\frac{c}{2b_p}\right)$$

Substituting and simplifying to make this driven by angle 1, the equation for angle 3 is finalized in the next equation.

$$\theta_3 = 2 \arcsin(\cos(\theta_1))$$

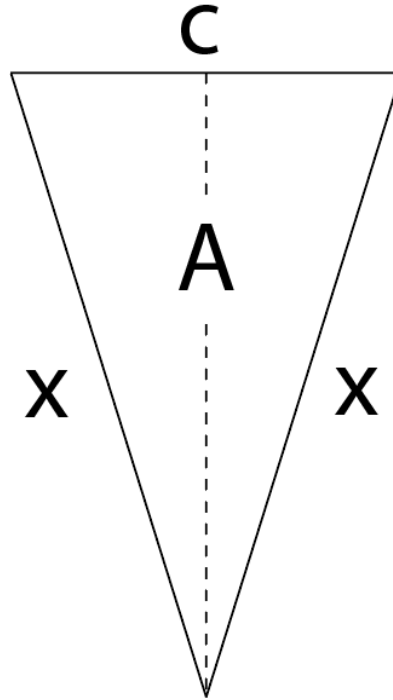
Next is angle 4.



The goal here is to define angle four by subtracting gamma from angle t. "C" is the same c from the angle 3 calculations, and A is the line bisecting the isosceles triangle with dimensions of x and c.

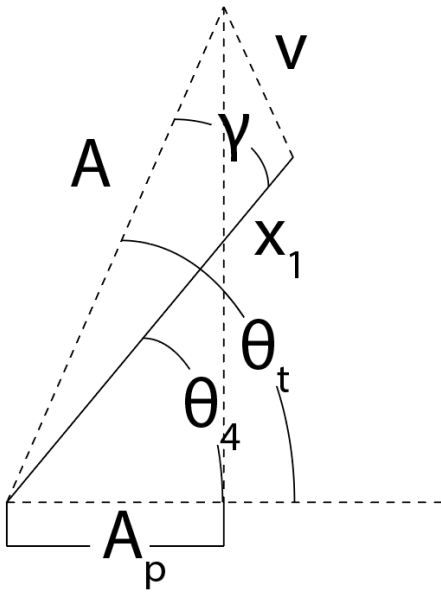
"V" is the line bisecting the isosceles triangle made of c and b. This results in the following equation.

$$v = \sqrt{b^2 - \frac{c^2}{4}}$$

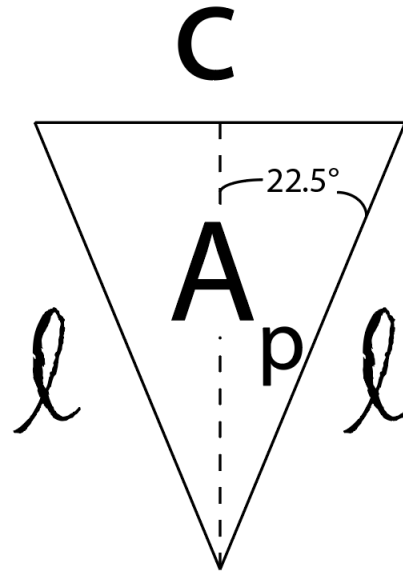


A diagram is shown above to be clear about the derivation of A. As a result, the relationship is shown below:

$$A = \sqrt{x^2 - \frac{c^2}{4}}$$



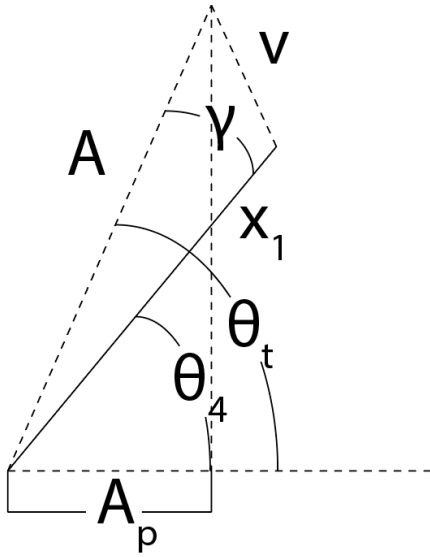
“Ap” is the projection of A onto the horizontal. However, it was defined using an isosceles triangle made of l and c used when finding angle 3. The image from that definition is reproduced below with the proper alterations.



As a result, the definition of Ap is as follows.

$$A_p = \frac{c}{2 \tan(22.5^\circ)}$$

Using the definitions of A and Ap, we can find angle t as a simple arccosine.



$$\theta_t = \arccos\left(\frac{A_p}{A}\right)$$

Using the correct substitution schemes and simplifying to make angle t driven by angle 1, we get the following formula.

$$\theta_t = \arccos\left(\frac{1}{\cos(22.5^\circ)\cos(\theta_1)}\right)$$

Finally, gamma can be defined by using the law of cosines and isolating for gamma.

$$\gamma = \arccos\left(\frac{A^2 + x_1^2 - v^2}{2Ax_1}\right)$$

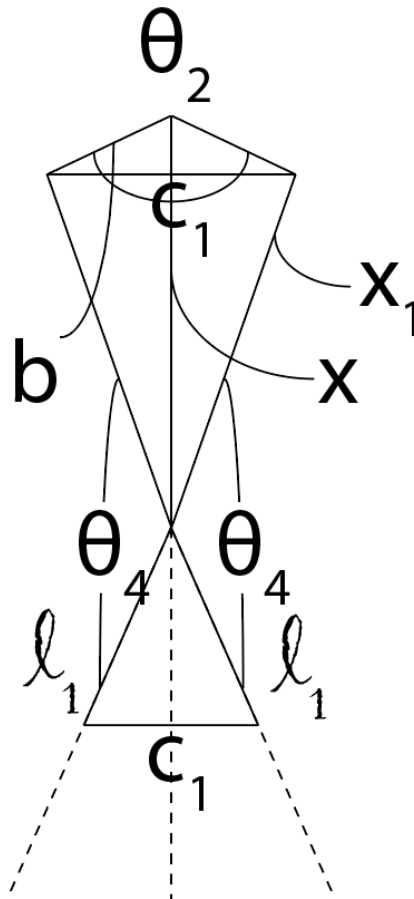
When all the correct substitution schemes are followed and simplified to make this equation driven by theta one, we get the following.

$$\gamma = \arccos\left(\frac{1}{\cos(22.5^\circ)\cos(\theta_1)}\right)$$

Now all that is left to define angle 4 is to define angle t and subtract gamma from it to get the following formula.

$$\theta_4 = \arccos\left(\frac{1}{\cos(22.5^\circ)\cos(\theta_1)}\right) - \arccos\left(\frac{1}{\cos(22.5^\circ)\cos(\theta_1)}\right)$$

Last is angle 2.



Like stated at the very start, angle 2 and 3 have very similar setups for their derivations, except angle 2 is much easier to derive. Dimension “b” is already at a right angle to the axis of the 45 degree fold so no extra steps are required except defining c and l which are now denominated by their subscript.

Like in the scenario with angle 3, l1 is the projection of x1 onto the dotted lines which is defined as follows.

$$l_1 = x_1 \cos(\theta_4)$$

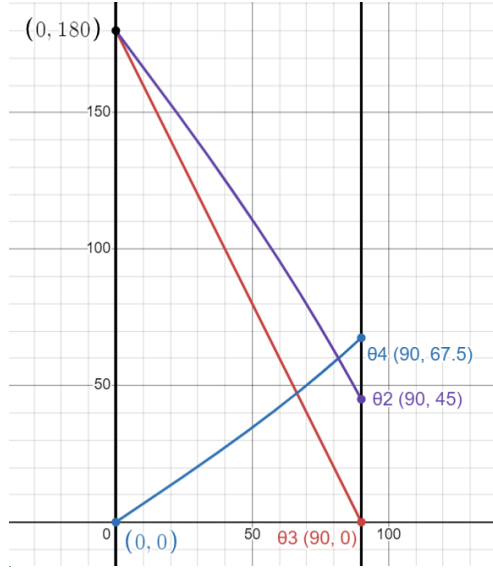
This allows us to define c1.

$$c_1 = 2l_1 \sin(22.5^\circ)$$

Finally, angle 3 is defined as follows. It is left in the form containing angle 4 to conserve space.

$$\theta_2 = 2\arcsin(\cos(\theta_4))$$

The accuracy of these equations were double-checked by comparing their limits to their predetermined values.

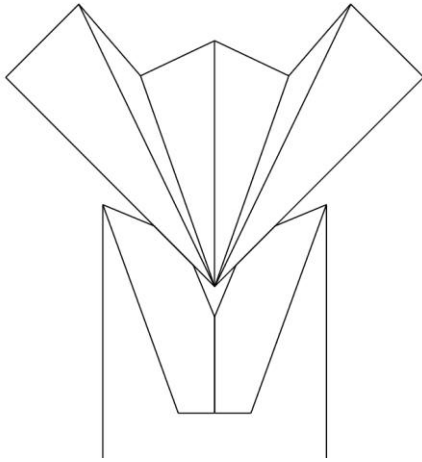


Graphed in desmos, the start and end positions represent open and closed. The values match the ones defined at the start of this derivation, confirming the accuracy of the equations.

3.3.2 Guide Rail Design

Using the mathematical equations derived above, we have the ability to know exactly where the array is in space as it opens circularly. This allowed us to actualize a guide rail design that would sit at the top of the supporting tower. The idea is that these static guide rails aid in the opening and closing of the array, and hold the array when the array is open or close.

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Above shows a sectioned view of the tower with the array sitting on top of it. It is called a “multi-stage” design because the guide rails guide the array to different levels of opening in a finite number of stages.

It is to be noted that this was the design modeled in the duration of this research, but it is not the most ideal known design.

The ideal guide rail design would take the multi-stage guide rail into a continuous and smooth guide rail design. This would be more ideal mostly for the top part of the design.

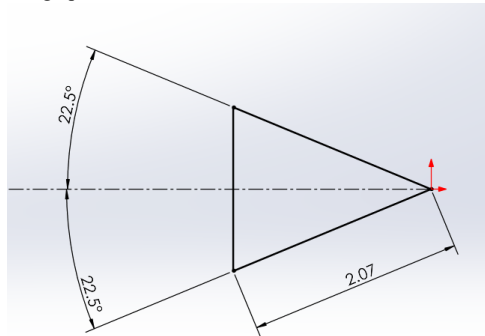


Looking at the fully modeled version of the multi-stage design above, it can be seen that the guide rails have gaps where panels poke through as they open. If this guide rail system is surrounded by a perimeter tower wall, the entire design becomes obsolete because parts of the array will rest on the top part of the wall, rather than the guide rail stages. However, this is necessary because the multi-stage design must end with one final stage that is just a flat plane to hold the array in its open state.

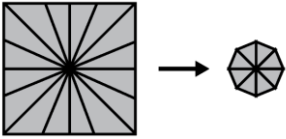
The only barrier for us to achieve the ideal smooth design was Solidworks. With little time left in the program, it was not figured out how to essentially turn the multi-stage design into an infinite multi-stage design in the modeling program. However, the shape itself is definitely feasible.

3.3.2.1 Zero Thickness Version

This section will go over exactly how the mathematical model aided in creating the guide rail design. It was done without thickness first to make the design process easier to break down.



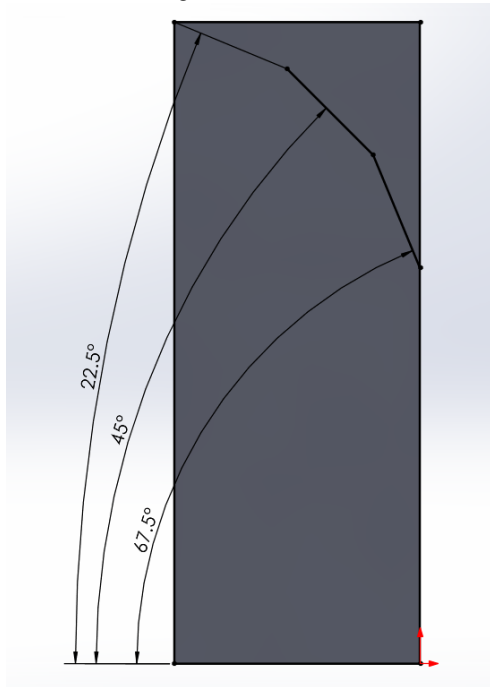
Shown above is the base shape of the main extrusion that is shaped into the guide rail. It is a triangular shape that mimics the space between papers from a top-down view, as shown below.



As we know from before, these sections have 45 degrees each, making the base shape an isosceles triangle with its defining angle being 45 degrees.

Its equal side lengths are dimension "b" from the mathematical derivations section. It is extruded to the value of dimension "x" because that is the point at which "a" panels taper in when folded. These sections of the array will not touch any guide rails, so the guide rail does not have to be as tall as the "a" panels.

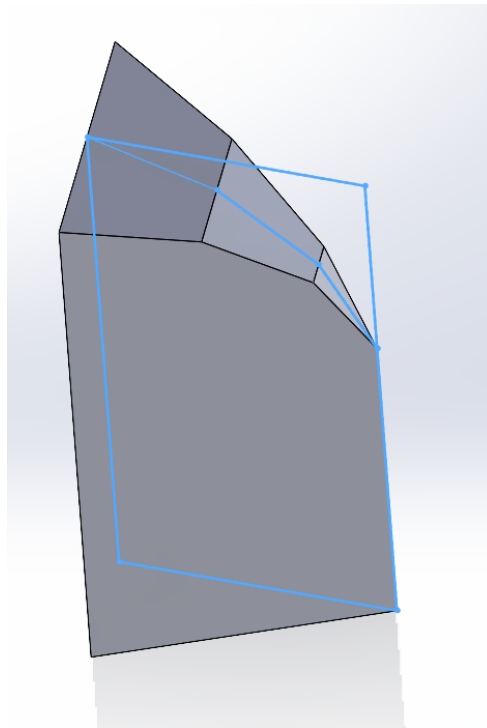
Next the stages must be defined.



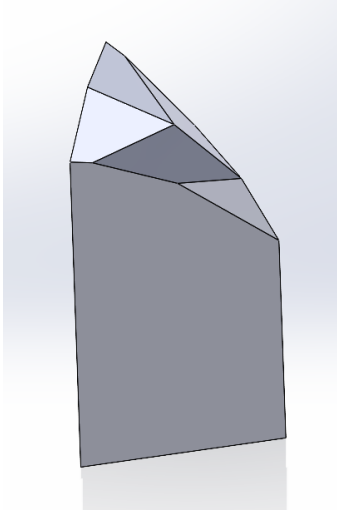
The plane used to cut out the side profile of the guide rail includes the line bisecting the 45-degree angle

in the base shape. This is because each guide rail will mold the resultant shapes of angle 2, the 45-degree fold, as the array moves. The resultant molded shapes will originate from the axis of the 45-degree folds, extruding out symmetrically in opposite directions.

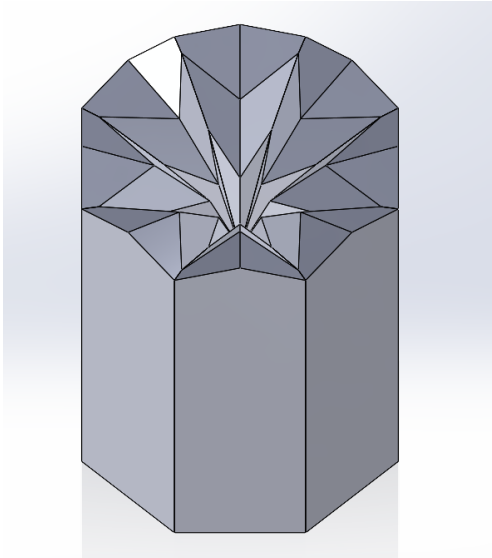
Since the multi-stage design is not ideal, there is not an easily definable criteria for the "best" design. However, it was determined that having equal lengths for the stages would keep forces of friction more consistent and minimized, and that the angle reductions should also be at a constant rate for the same reasons. This results in the side profile shown above. It would technically be a 5 stage design. 2 of the stages are open and closed, and the other three are in between.



From here, each stage's final cut outs can be made by using the relation between angle 1 and angle 2. This results in the shape shown below.



When this is assembled in an assembly, it results in the image below.

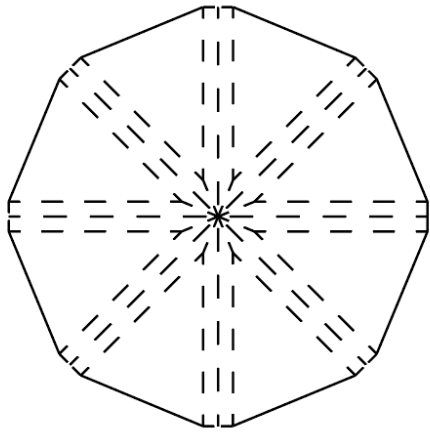


The guide rails align perfectly to mimic the array at different stages of opening, as desired. This would work for a zero thickness model as stated in the beginning. It is to be noted that the beginning stage is clearly not possible

because the zero thickness essentially turns the closed stage into nothing. Making this work for the real thick array is a matter of dealing with this issue.

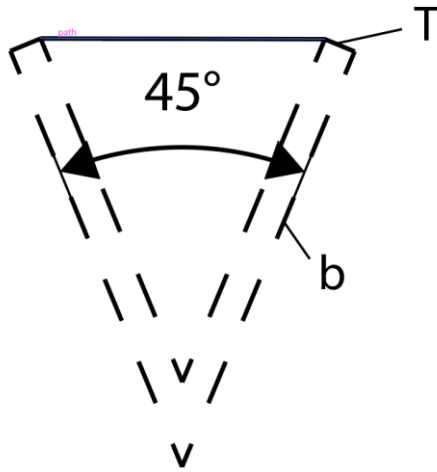
3.3.2.2 Accommodated for Thickness

The guide rails are accommodated for thickness by taking a look at the thickness and space of the array when it is closed from a top down view.

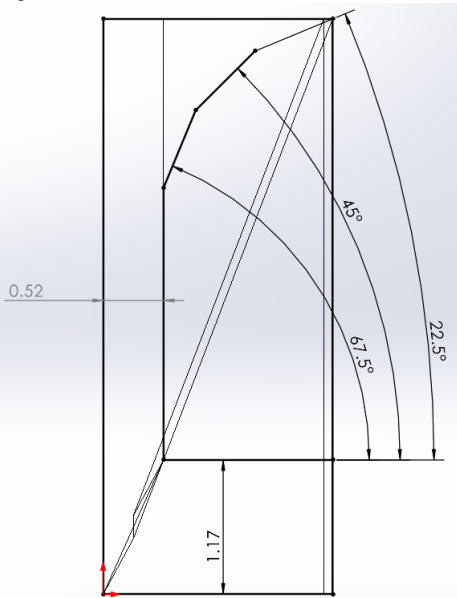


The image reproduced above shows that the shape that would surround the array isn't an octagon, but really a 16 sided polygon.

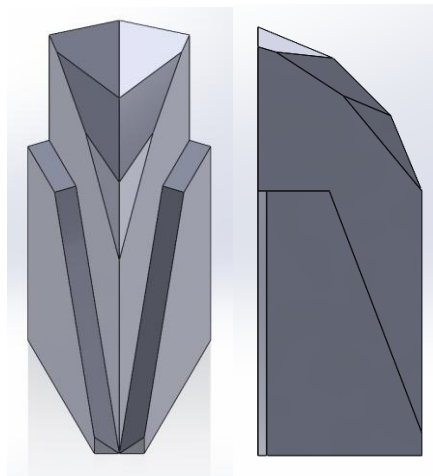
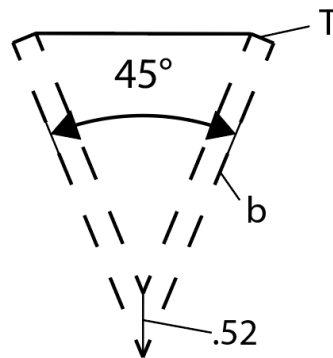
This shape above allows us to change our base shape for the guide rails from a triangle to a quadrilateral shaped like a triangle, as shown below. "T" is a new variable that refers to the thickness of the array. All other dimensions have been previously defined.



Using this base shape, it is once again extruded to a height of "x", and the final cuts are done the same way using the derived formulas, however, the side profile changes.



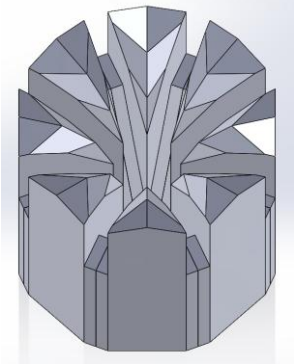
The main change is that the closed stage, which is represented by the main vertical line at the start of the guide rail line, is displaced to accommodate for the thickness. Dimension 1.17 can really be any value, however it was chosen as the value here. This is because there needs to be a through hole through the center of the guide rail system to allow for connections to the main power source and the array. The important dimension is .52. This was calculated by the image below.



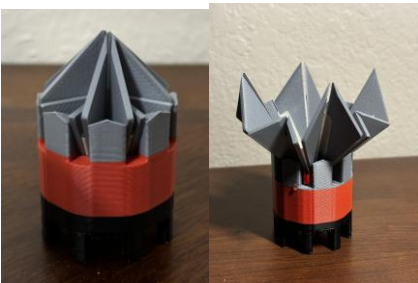
Shown above is the finished guide rail with thickness accommodated. One last thing to note is that there is a cut made to the sides of the rail to level the sides to the beginning of the first opening stage that is not

folded. Those two extrusions are actually just a direct mold of the array when folded. The reasoning for this is similar to the explanation of how adding a perimeter wall around this rail would make the stages obsolete. If this cut is not made, the first stage would take over the other stages.

This system put together results in our final “working” but not ideal guide rail system.



This was printed as a proof of concept. It performed as expected and images of the test are shown below.



4. Data Analysis

4.1 Methodology

Data analysis is composed of two major goals, to analyze change in corn crop yield after implementing various agrivoltaic conditions, and comparing the energy production of the agrivoltaics. |

Homer Pro software is a simulation software that allows the modelling microgrid of a set of time and can show scenarios with different systems and equipment in a scenario. Using the specifications of the BSG 150/69F Brite Solar Greenhouse Solar Glass, as well as the designed solar array factors such as the the rated capacity, nominal operating temperature, temperature coefficient, and efficiency were input into HomerPro to produce the annual total power production operating cost and life cycles. The rated capacity of our designed system array using Brite Solar panels was calculated by taking the kilowatt per square meter ratio and multiplying it by panel area. After calculations, the rated capacity came out to be 0.379kW.

To compare the efficiency of our scenario using Brite Solar Transparent Panels two more common brands of solar panels were modeled in HomerPro. The Trinasolar Vertex N NEG19.RC.20 with a thickness of 30mm and the Trinasolar Vertex N NEG20C.20 with a thickness of 33mm were used in two additional HomerPro models. The same factors such as rated capacity, nominal operating temperature, temperature coefficient, and efficiency were used in the model. To ensure a more fair and accurate comparison between the different panel types, the rated capacity of each panel type was determined using our designed array area ratio as the designed solar array system was used. KW rated capacity of TrinaSolar Vertex N NEG19RC.20 30mm was 1.27 and the TrinaSolar Vertex N NEG20C.20 33mm was 1.22 kW. From the three different panel scenarios that were input into HomerPro, operating cost, total annual production, and maximum output results were determined.

4.1.2

Crop growth data analysis was modeled using Agricultural Production Systems sIMulation (APSIM). This program allows simulations and agriculture systems modeling. This advanced system has the capability to consider factors such as crop type, soil fertility, livestock rotation, climate change, and management style impact. To create a model that reflected our scenario weather, location, harvest, and crop parameters were adjusted to our needs.

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APSIM offers example models of different crops which have basic parameters already set to allow for more accurate modeling based on their research and programming. As a foundation for our models the Maize file example was used in every simulation. Since no program has a model scenario set with agrivoltaics we customized APSIM weather MET files to best portray the various climate scenarios we wanted to consider. Weather data from West Palm Beach, Florida was gathered from the National Solar Radiation Database and local weather station data _____. Daily radiation and vapor pressure data was found through the NSRDB, while maximum and minimum temperature and precipitation data was found using _____. All required data was then compiled into the APSIM weather.met file format.

After all weather data from 2020-2022 was assembled, eleven different scenarios were modeled over the course of the three years. The first scenario had no alterations to weather or climate data, this stood as a model for a field scenario with no agrivoltaics. The second scenario was with our agrivoltaic system. Since the transparency level recommended by Brite Solar for growing corn was 69.35%, we edited the radiation data to be reduced by 30.65%. The following four scenarios also only had changes in the radiation factor. We did scenarios for 60%, 50%, 40%, and 30% agrivoltaic transparency, and reduced the radiation by each correlating factor. Models with increasing reduction in transparency were done to observe and compare the change in crop yield with different transparency levels. The lowest transparency was 30% as that was determined to be the approximate lowest possible level where crop yield reached zero. Five other scenarios were modeled as well, they replicated the transparency levels of the second through sixth scenario. However, there was an added reduction in the maximum and minimum temperatures by two degrees in order to consider agrivoltaic shading impact on crop growth. To account for agrivoltaic coverage of different panel types radiation and temperature weather data were customized in various ways throughout the eleven different scenarios. Precipitation and vapor pressure were never altered.

For every scenario, the only other data changed from the foundational example maize files were sowing

season and crop type. As we wanted the analysis of our array design to be in the context of American agriculture, we changed the sowing season to be from April 1st to May 10th instead of January through February which it was initially set to. This resulted in most models' corn crops being harvested around August, which is within the common season for farmers within the United States to harvest corn. Furthermore, for all scenarios we used Dekalb XL82 as the maize variety because it was the most optimal option available to us since Dekalb is a variety widely found in the United States.

After setting all parameters and troubleshooting any issues, the APSIM model produced the total yield, grain weight, and harvest date of each scenario which was used to solve the percent reduction in crop yield based on the maximum yield determined by the scenario with no agrivoltaic factors implemented.

T Test
4.2 Results

Panel Type	Transparent Pv (69.35%)	Trina Solar 30mm	Trina Solar 33mm
Individual Panel Area of One System (m^2)	2.177	2.701	2.83
Rated Capacity (kW)	0.379	1.27	1.22
Annual Total Prod. (kWh)	568	2157	2069

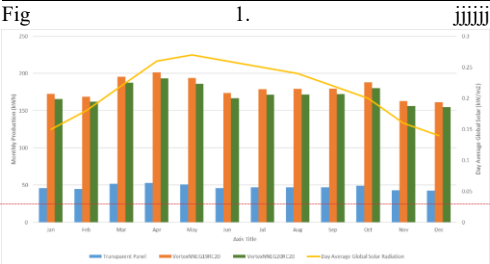


Fig 2.

Year	Transp %	Temp Red (oC)	Yield (g/m²)	Grain Wt (g/m²)	Harvest	% Decrease
2022	1	0	9167.584	916.758	22-Jul	0

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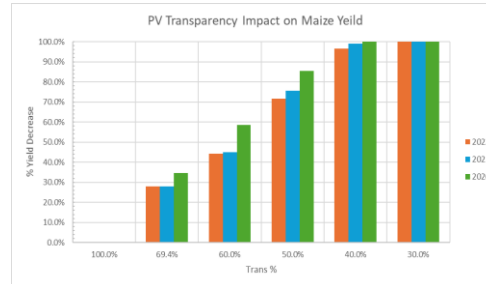
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2022	0.6935	0	6593.421	659.342	22-Jul	0.2807
2022	0.6	0	5113.942	511.394	22-Jul	0.4422
2022	0.5	0	2595.094	259.509	22-Jul	0.7169
2022	0.4	0	310.581	31.058	22-Jul	0.9661
2022	0.3	0	0	0	22-Jul	1
2022	0.6935	2	6869.066	686.907	31-Jul	0.2507
2022	0.6	2	5927.482	592.748	31-Jul	0.3534
2022	0.5	2	3656.682	365.668	31-Jul	0.6011
2022	0.4	2	1108.825	110.882	31-Jul	0.879
2022	0.3	2	0	0	31-Jul	1
2021	1	0	9923.322	992.332	28-Jul	0
2021	0.6935	0	7149.216	714.922	28-Jul	0.2796
2021	0.6	0	5459.544	545.954	28-Jul	0.4498
2021	0.5	0	2424.344	242.434	28-Jul	0.7557
2021	0.4	0	95.083	9.508	28-Jul	0.9904
2021	0.3	0	0	0	28-Jul	1
2021	0.6935	2	6458.083	645.808	8-Aug	0.3492
2021	0.6	2	5224.745	522.475	8-Aug	0.4735
2021	0.5	2	2915.283	291.528	8-Aug	0.7062
2021	0.4	2	586.844	58.684	8-Aug	0.9408
2021	0.3	2	0	0	8-Aug	1
2020	1	0	9305.608	930.561	4-Aug	0
2020	0.6935	0	6088.56	608.856	4-Aug	0.3457
2020	0.6	0	3845.557	384.556	4-Aug	0.5867
2020	0.5	0	1364.681	136.468	4-Aug	0.8533
2020	0.4	0	0	0	4-Aug	1
2020	0.3	0	0	0	4-Aug	1
2020	0.3	0	0	0	4-Aug	1
2020	0.6935	2	6843.287	684.329	16-Aug	0.2646
2020	0.6	2	5272.844	527.284	16-Aug	0.4337
2020	0.5	2	2559.556	255.956	16-Aug	0.7249
2020	0.4	2	244.601	24.46	16-Aug	0.9737
2020	0.3	2	0	0	16-Aug	1

Fig

3.



Fig

4.

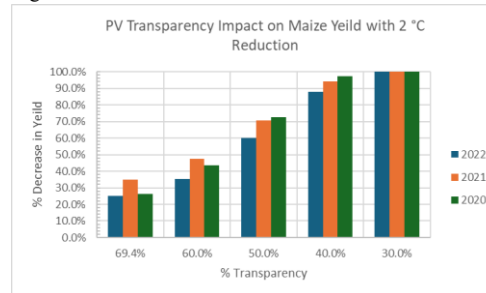


Fig 5.

5. Discussion

The APSIM model results show an inverse relationship between a percent decrease in yield and panel transparency. The model supports logical hypotheses that as the transparency decreases, more radiation is absorbed by the solar panels to create energy that less is available for the plants to use for photosynthesis. Our results show a clear tradeoff, to gain more efficiency in solar energy production you sacrifice plant growth and total yield. The simulations representing our proposed agrivoltaic desing with a panel transparency level of 69.35% without temperature reduction show a 27.96-34.57% decrease in yield (Fig 3.). The 69.35% transparency scenario with temperature reduction shows a 25.07-34.92% decrease in yield (Fig 3.). The transparency level where the yield of maize reached zero was found to be around 30%. As the transparency decreases in increments from 69.35%, a

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steep trendline is shown. Already at 40% transparency the yield has decreased by nearly 100%.

To account for additional factors that may effect yeild besides radiation, a set of simulations was done with a 2 degree decrease in both maximum and minimum daily temperature. The study ____ by ____ suggests that areas with shading may have a maximum two degree decrease in ambient/ground temperatures. Throughout the literature review preceding this research, it was never determined wether temperature in the soil or ambient environment has a larger impact on plant growth. This distinction matters as air temperature is measured in the shade, and would mean changes in temperature caused from overhead cover would not need to be accounted for. Furthermore, there was no evidence available regarding the impact of shading or soil temperature reduction on agricultural yield. To account for the possibility that there is a temperature reduction due to shading, a variation of the model scenarios was done with a 2-degree temperature reduction for all scenarios. When comparing the simulations with no temperature reduction versus with a temperature reduction, one trend is that the percent decrease in yield is slightly lower when a temperature reduction is added. This may suggest that a lower temperature is beneficial for the growth of maize. If evidence arises that agrivoltaic coverage does provide some temperature reduction, integrating these systems may actually be beneficial to plant growth.

The HomerPro modelling shows the annual total production and rated capacity of three different solar panel models, the Brite Solar BSG 150/69F transparent PV modeled for our array, the Trina Solar Vertex N NEG19RC.20, and the TrinaSolar Vertex N NEG20C.20. We chose to use Trina Solar panels as a comparison because Trina Solar is a widespread brand that manufactures a more generic type of panel. The annual total production of the transparent PV in our array is only 568 kWh (Fig 1.). This is significantly lower the annual total production of the Trina Solar panels which are all over 2000 kWh. The transparent panels have a significantly lower energy production efficiency and so seem inferior to generic solar panels in this sense. However, it is important to consider the

tradeoffs of transparent solar panels. Although not as much energy is being produced, you are able to grow crops with little effect. This research can be used to help farmers and researchers begin to guage wether integrating transparent agrivoltaics would be economical to them based on the modeled potential energy production. In a large scale farm or ranch with a large amount of panels, 568kWh of annual total energy production per panel would compound into a substantial amount of solar energy.

To further understand the potential application of this research caluculations for an example farm were made. The APSIM models were based on a field size of 1ha or 10,000 meters squared in West Palm Beach, Florida. To account for harvesting or spraying vehicles the most common tractors and machines in maize agriculture was researched. The spacing needed between agrivoltaic foundations would need to be a range of roughly 9.5-14.5 meters. Our design is 3.048 meters by 3.048 meters. In a 10,000 square meter field, an optimized layout would have 148 agrivoltaic systems. This would be 24.26% panel coverage and 75.74% open field or natural corn growth. Based on HomerPro optimization simulation, 148 arrays in a field in West Palm Beach, Florida would produce 64.06 MWh of solar energy per year. The maize crop yield under the agrivoltaics would only be reduced by 25-35% and all other crops would grow as expected in an unaltered environment. The economic feasibility of our design and this scenario can be further researched in the future.

During the 9 week's research period, lots of impactful progress was made regarding the research and investigation of the feasibility of transparent agrivoltaics in farms and ranches. However, some aspects of the research were not perfect and if continued further could be greatly improved. First, our research focused on agrivoltaics and transparent photovoltaics, two topics that are still in a developmental phase and do not have extensive research or historical studies available. In order to answer the questions we had, we decided to use programs and software such as HomerPro and APSIM that were not specifically made for the applications for which we intended to use them. For HomerPro, we modified preexisting generic photovoltaic panels with

the specifications of transparent photovoltaics. APSIM as well, has no set parameter for agrivoltaic coverage and so through some research, we decided that modifying radiation and temperature would likely give us the closest result to the real scenario. These respected softwares and models have gone through rigorous testing and development in order to offer researchers and users accurate results. Our flexible use of these technologies may have jeopardized some accuracy of our results.

Furthermore, APSIM is a program used for a very niche audience. We had less than two months to familiarize ourselves with this very complex program. APSIM was developed and is managed in Australia and New Zealand. Parameters set in the example files of this software were set based on climate and agricultural research done in these regions. In order for our research to be more widely applicable we focused our study on West Palm Beach, Florida which has a very different climate and environmental characteristics than Australia or New Zealand. For the simulations we only customized weather, sowing season, and maize variety. However, there are dozens of other parameters available to customize. The actual agricultural yield that would occur in these scenarios located in West Palm Beach may be very different than the results of our model.

For our custom weather files, we used data over the years 2020-2022 from the National Solar Radiation Database (NSRDB) and _____ which pulls from the local weather station in West Palm Beach. The NSRDB does not use recorded data but creates the data through simulations. In order to verify if the data used from the NSRDB was accurate and suitable for use, we performed a T-Test between recorded data and NSRDB data for precipitation and temperature. The T-test passed for temperature but failed for precipitation. We decided to still use the NSRDB data for radiation and vapor pressure data.

Though we decided to use APSIM modeling as a way to gauge agrivoltaic coverage effect on crop yield when communicating with Brite Solar the transparent PV manufacturer about panel transparency suitable for the corn crop, they ensured us that the 69.35% transparent PV is suitable for corn crop growth. They also emphasized that the transparency levels of their

panels are all customized to the specific needs of different plant varieties and had been researched and studied by a third party. Though they did not disclose with us the evidence or research data, it can be assumed according to Brite Solar that photovoltaics with a 69.35% transparency level may have no impact on corn or maize cultivation.

6. Conclusion

As climate change becomes a more pressing issue, industry, governments, and communities, are shifting their focus onto pursuing more sustainable practices. Transitioning from fossil fuels to renewable energy is one of the biggest challenges but most important in reaching Net Zero and sustainable development goals. Agrivoltaics is a developing technology merging agriculture and renewable energy. Growth of this sector would greatly increase the amount of renewable energy produced. The implementation of this technology has been slow to develop due to some obstacles. We thought that adapting transparent PVs to agrivoltaics would greatly combat these issues. Our research focused on the feasibility of transparent agrivoltaics in farms and ranches. We addressed this by first designing an array and using simulations to analyze the impact of agrivoltaics on agriculture.

. We were able to develop a prototype design of a transparent agrivoltaic array that is collapsable and allows for full functionality of equipment. The data analysis showed the decrease in maize yield when implementing different levels of transparent agrivoltaics and a comparison of solar energy production between transparent and generic agrivoltaics.

Our research helps farmers and researchers gauge new possible ways to implement agrivoltaics and the potential tradeoffs which may occur. Agrivoltaics and transparent agrivoltaics are still developing field and require much more field research. Agrivoltaic effect on crop growth should be further researched. The design for the array should also be further developed and tested in the future. Additionally, an economic analysis considering the cost and benefit of all factors of implementing transparent agrivoltaics should be done. If

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this topic is explored further it would make farmers more interested and hopefully provide convincing evidence of why it should be implemented and what benefits it has.

7. References

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